

LOCKHEED MARTIN F-22 RAPTOR (US)

- ***Development cost:*** **$67 billion (₹5.49 lakh crore)**
- ***Entered service in:*** 2005
- ***Features:*** Highly manoeuverable supersonic, multi-role fighter jet with stealth and supercruise capabilities. Clipped diamond-like delta wings, angular fuselage. Top speed greater than Mach 1.8 (2,222 kmph) without afterburners
- ***Armament:*** Six AIM-120 AMRAAMs and two AIM-9 Sidewinder air-to-air missiles; for ground targets, it can internally carry two 1,000-pound GBU-32 JDAM bombs



is V-shaped. It has an internal weapons bay and other stealth features
- An internal payload of 1,500 kg, external payload of 5,500 kg, with 6,500 kg of internal fuel
- Supercruise ability, data fusion and multi-sensor integration with AESA (active electronically scanned array) radars

ARMAMENT: BrahMos-NG air-to-ground missiles, Astra air-to-air missiles, anti-tank missiles, Rudram air-to-ground anti-radiation missiles, laser-guided bombs and precision munitions

CHENGDU J-20 MIGHTY DRAGON (CHINA)



- ***Development Cost:*** **$30 bn (₹2.46 lakh crore)**
- ***Entered service in:*** 2017

Features:
- J-20's design focuses on stealth, with features that reduce its radar cross-section. Integrated avionic suite, KLJ-5 active electronically scanned array radar. Top speed of approx. Mach 2 (2,470 kmph)
- ***Armament:*** The main weapons bay can house up to six long-range missiles or bombs, for both air-to-air and precision-guided munitions; side bays carry short-range air-to-air missiles, such as the PL-10E

SUKHOI SU-57 FELON (RUSSIA)



- ***Development cost:*** **$15 billion (₹1.23 lakh crore)**
- ***Entered service in:*** 2020
- ***Features:*** Twin-engine, stealth multirole fighter with supercruise, integrated avionics, active phased array radar and advanced flight control system. A powerful propulsion system, equipped with Izdeliye 117 or AL-41F1 augmented engines power it to speeds of up to Mach 2
- ***Armament:*** Four beyond visual range missiles (R-77M), two short-range missiles (R74M2), anti-radiation missiles, precision-guided bombs and Kh-38M air-to-ground missile

technology that reduces the heat signature of the engine and special radars to minimise emissions that enemy radars can pick up. Tied to stealth capability is the internal storage of smart weapons and the ability to supercruise—the sustained supersonic flight of a jet without using afterburners, the component in jet engines used to increase thrust during take-off and supersonic flight—thus making it truly trail-less. The ADA claims to have developed similar technology indigenously. The AMCA will be based on the concept of 'first look, first kill' where an AMCA pilot would see an enemy plane first, fire a missile and destroy it before the latter can react. Typical qualities of a fifth gen fighter like low observability and 'first look, first kill' capabilities are unavailable in most front-rank fighter jets. In the IAF inventory, not even the Rafales have them. The AMCA will be designed to perform a variety of missions, including air supremacy, ground strike missions, suppression of enemy air defenses (SEAD) and electronic warfare.

AMCA designers say that the jet will have a top speed of nearly 2,600 kilometres per hour (Mach 2.15), as well as a combat range of 1,620 km. The fighter will be equipped with a 23 mm cannon and 14 hard points to carry weaponry weighing 6,500 kg. The design is intended for multi-role missions, and to tackle both air-to-air and air-to-ground operations with equal lethality. It also features serpentine air intakes, extensive use of radar-dampening materials, a conformal antenna (embedded in the fuselage, rather than sticking out from it, as in non-stealth jets), and an internal weapons bay—all essential to provide a very low radar signature.

A defence source says that the AMCA will carry BrahMos-NG (next